OSPF Setup

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Introduction

Open Shortest Path First (OSPF) is a dynamic link-state routing protocol used to determine the most efficient path for data transmission across a network. It uses Dijkstra's Shortest Path First (SPF) algorithm to calculate the best routes based on link costs. In this practical setup, three routers, PCs, and a server are configured with appropriate IP addresses and OSPF settings to establish communication between different networks and demonstrate dynamic routing.

OSPF

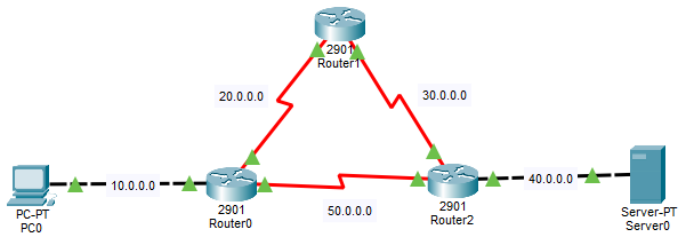
OSPF (Open Shortest Path First) is a dynamic link-state routing protocol that finds the most efficient path for data within a network. Instead of counting hops, it uses Dijkstra's SPF algorithm to calculate the best routes based on interface bandwidth. Every router maintains a synchronized map of the entire network topology to make smart, real-time routing decisions.

The main advantages of OSPF are its fast convergence, scalability, and loop-free design. It adapts to network changes instantly by sending targeted updates rather than full routing tables. By dividing large networks into logical areas, it easily scales to massive infrastructures, supports equal-cost load balancing, and works seamlessly across multi-vendor hardware.

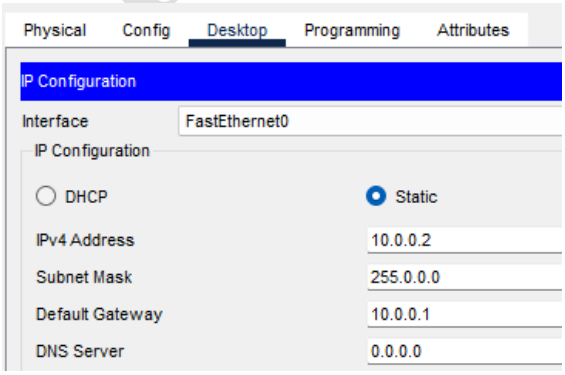
However, OSPF's complexity makes it resource-intensive and difficult to manage. Running the SPF algorithm requires high CPU and memory usage, which can strain older routers. It demands advanced networking expertise to configure properly, can cause bandwidth overhead if unstable links flood the network with updates, and is generally overkill for small networks.

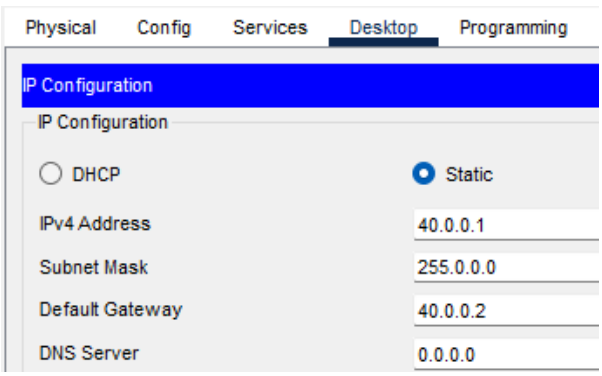
OSPF Setup

To setup OSPF protocol we will need some devices, PC, 3 routers, and a server as shown below:

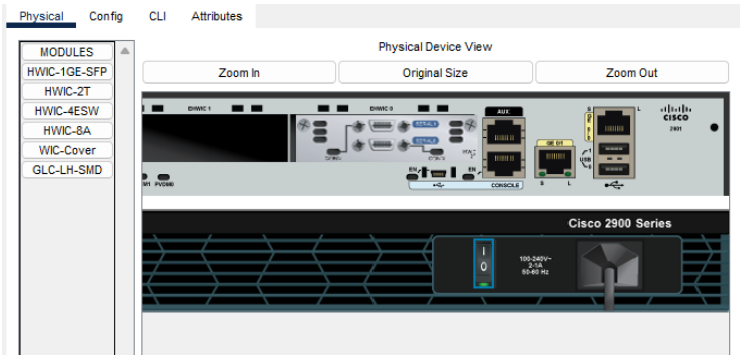


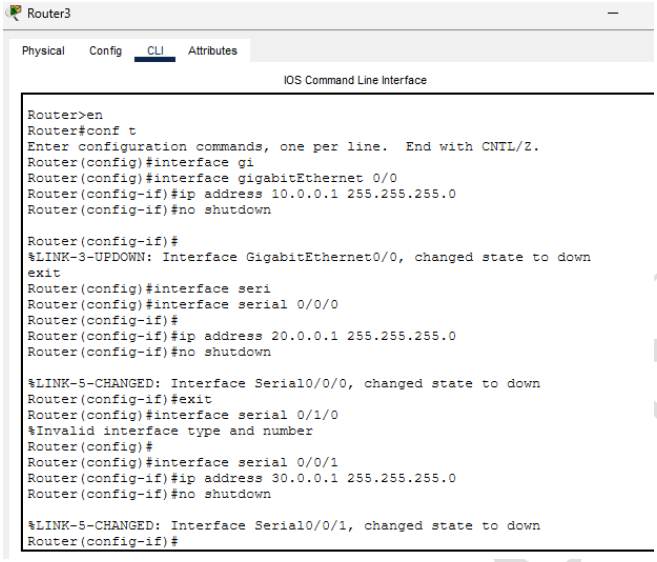
As we did previously with the setup of RIP, we need to assign IPs for particular network for them to communicate with each other. So, we need to configure the IP for each device according to the IP series assigned to the network.





For the IP configuration of router, we need to add the serial port for the routers to communicate between them through serial DCE.





After IP is configured for each device then they will be able to communicate with device neighboring them as they share same series IP, but for them to transfer over the network then we need to add hops for the packet to be transferred through each small network through routers, as we are seeing this for OSPF protocol we will add hops through OSPF for efficient packet transfer.

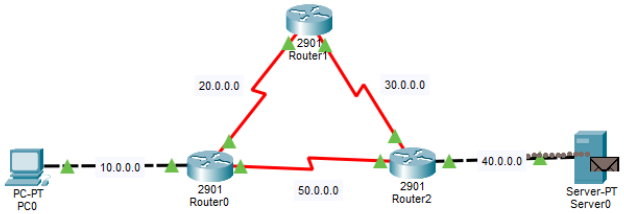
OSPF Configuration

To configure OSPF, we need to open CLI on router and enter following commands to enable OSPF protocol:

```
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 0.255.255.255
% Incomplete command.
Router(config-router)#network 10.0.0.0 0.255.255.255 area 0
Router(config-router)#network 20.0.0.0 0.255.255.255 area 0
Router(config-router)#network 30.0.0.0 0.255.255.255 area 0
Router(config-router)#end
```

The commands go as “router ospf 1” here we are enabling protocol i.e. ospf and the one signifies ID. Then the network command is for whatever the network IPs are connected to the router in this case there are 3 networks connected so we specified it as, but you will notice the subnet is reverse which is for reserving the IP series. After we are done with this with each router then we are able to send to packet over this particular network. Let’s see:

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Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Router0
	0.003	Router2
	0.004	Server0

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC0	Server0	ICMP		0.000	N	0

Conclusion

The OSPF configuration demonstrates how routers exchange network information and dynamically determine the best paths for packet transmission. After assigning IP addresses and configuring OSPF on each router, devices across different networks can communicate without manually adding routes. This practical helped in understanding link-state routing, network statements, wildcard masks, and the role of OSPF in providing efficient, scalable, and reliable network communication.

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